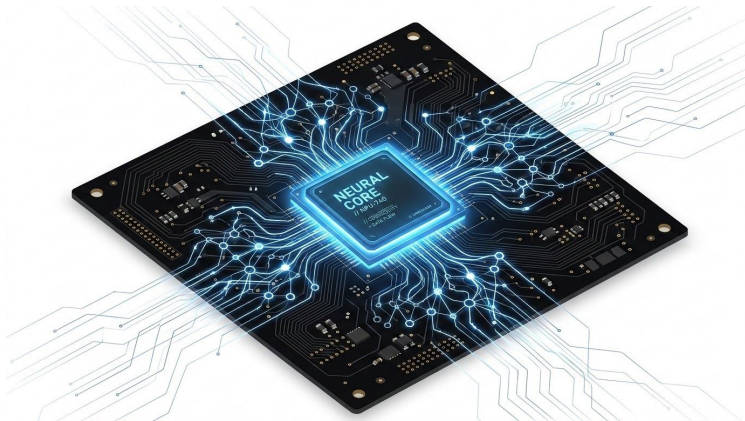


Lifecycle of Edge ML Systems

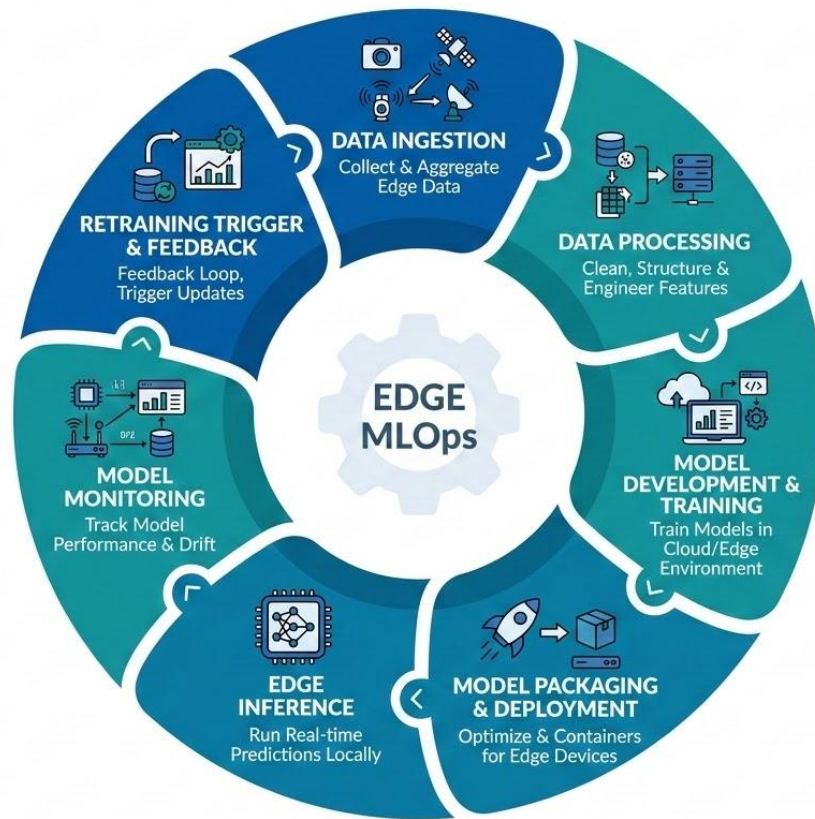
Postgraduate Program in Embedded Computing Systems — ISEP
From isolated inference to continuous real-world operation

Instructor: VMS | Runtime: LiteRT (Google Edge) [\[Web\]](#)



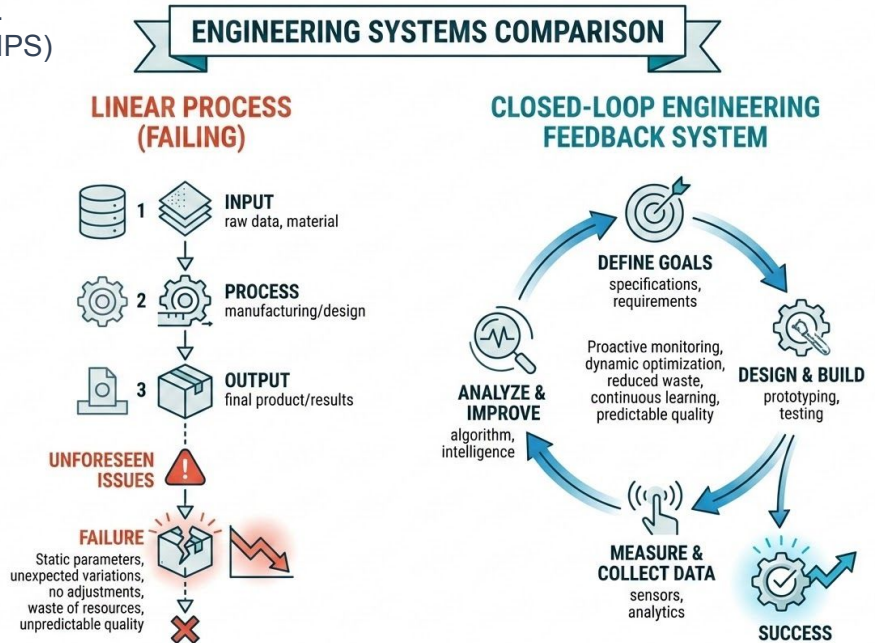
Syllabus & Core Topics

- Deployment does not end with the first inference
- Operational and statistical local monitoring
- Data variations (Data Drift & Dataset Shift) [Paper]
- Performance degradation & anomaly detection
- Model versioning with DVC [Tool] & registry with MLflow [Tool]
- Preventive and adaptive maintenance
- Edge security & adversarial robustness
- Continuous operation & fault tolerance



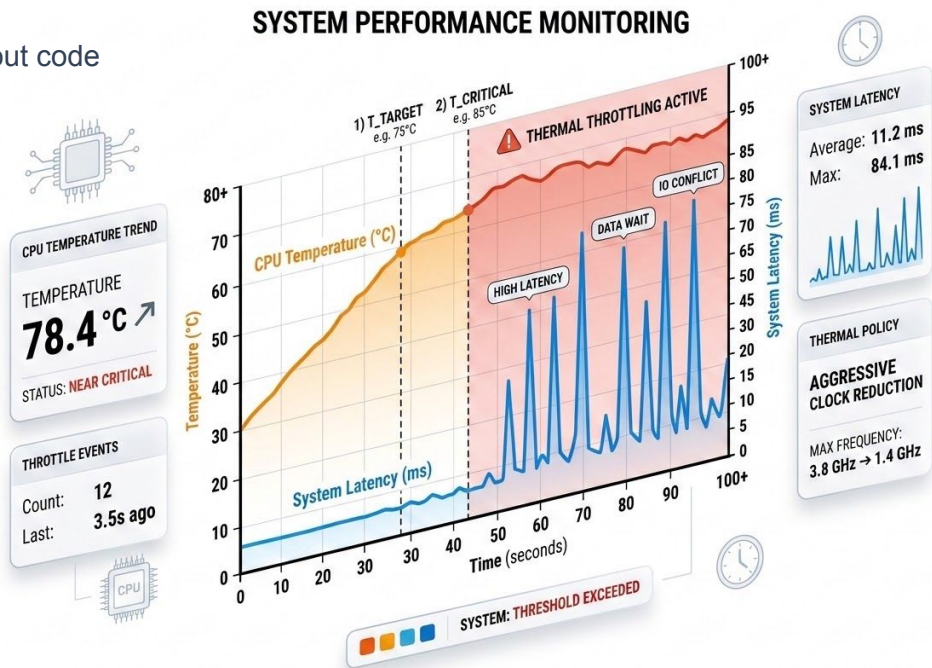
Deployment Is Just the Beginning

- The Linear Fallacy: Train -> Optimize -> Deploy -> Done.
- Industrial Reality: Dynamic closed-loop systems.
- Loud Failures vs. Silent Failures:
 - * OS and memory crashes are explicit and interceptable.
 - * Degraded neural nets still infer with high apparent confidence.
- Seminal Reading: Hidden Technical Debt in ML Systems (NeurIPS)
[\[Paper\]](#)
- Continuous maintenance accounts for the real system cost.



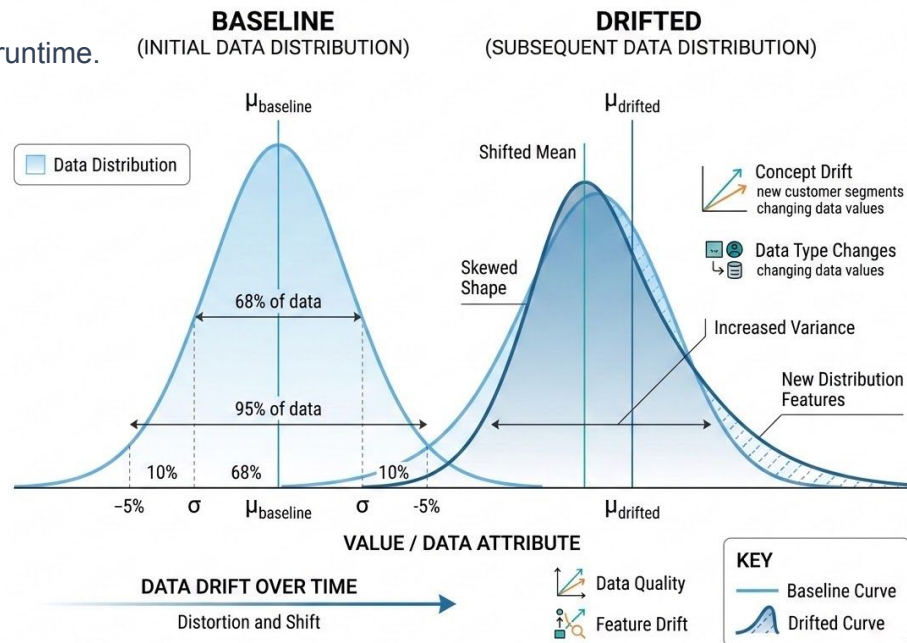
Infrastructure & Hardware Telemetry

- Local On-Device Telemetry:
 - * Inference latency (p95/p99 percentiles via Benchmark Tool) [\[Tool\]](#)
 - * SRAM utilization: Detecting tensor allocation memory leaks.
 - * SoC thermal monitoring & processor throttling.
 - * Voltage stability & power budget depletion.
- Operational Risk: Latency spikes and throughput drops without code changes.



Data Variations: Data Drift vs. Concept Drift

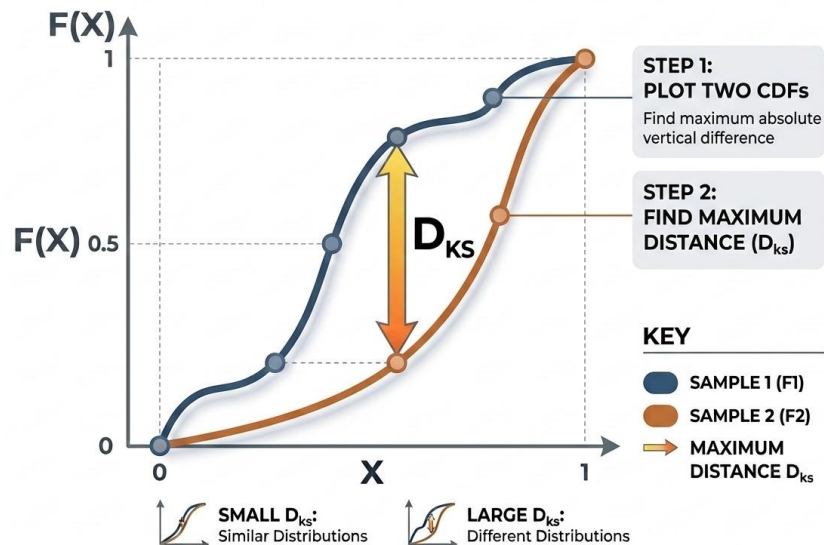
- Data Drift (Covariate Shift): Shift in input distribution $P(X)$ [\[Paper\]](#)
 - * Example: Lens dust accumulation, seasonal ambient lighting changes.
- Concept Drift: Shift in mapping relationship $P(Y|X)$.
 - * Example: Mechanical fatigue altering normal baseline vibration patterns.
- The Edge Constraint: Complete lack of ground-truth labels at runtime.



Unsupervised Drift & Anomaly Detection

- Lightweight Statistical Methods for Edge Nodes:
 - * Kolmogorov-Smirnov (KS) Test for continuous signals [\[Paper\]](#)
 - * Kullback-Leibler (KL) Divergence on activation histograms [\[Paper\]](#)
- Out-of-Distribution (OOD) Detection:
 - * Baseline OOD Detection (Hendrycks & Gimpel, ICLR) [\[Paper\]](#)
 - * Monitoring Softmax entropy decay and maximum probability drops.

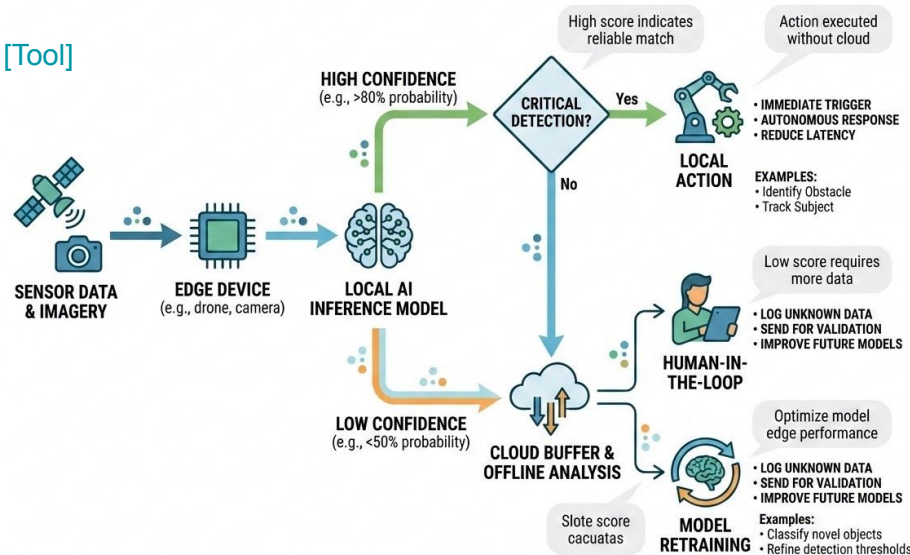
KOLMOGOROV-SMIRNOV (K-S) STATISTICAL TEST



Intelligent Telemetry & Selective Uploads

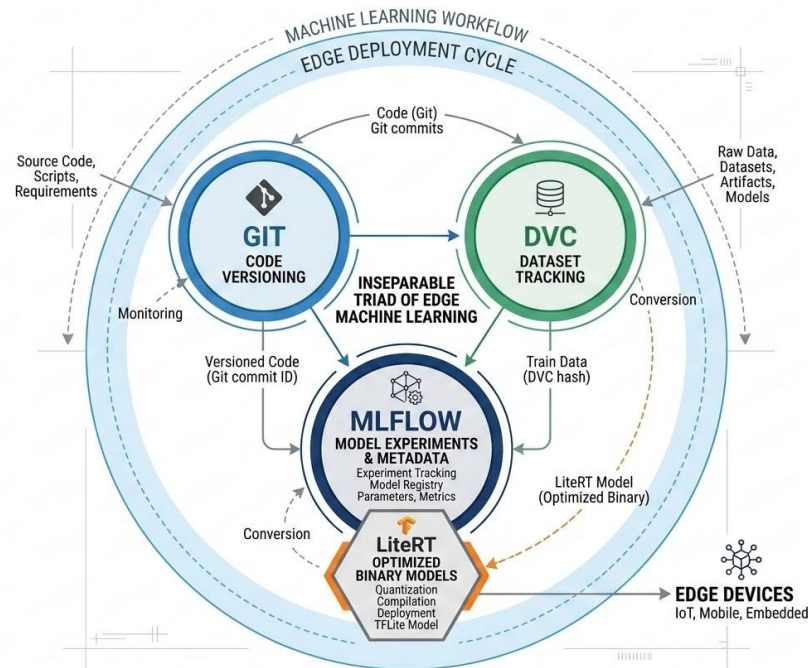
- Bandwidth Bottleneck: Streaming continuous raw sensor feeds is unfeasible.
- Selective Sampling Strategies:
 - * Uncertainty Trigger (Active Learning): Upload only samples with confidence $< 60\%$.
 - * Audit Sampling (Reservoir Sampling): Transmit 0.1% of random inferences.
 - * Intermediate embedding vectors instead of raw image data.
- Case Study: Predictive Maintenance C-MAPSS Dataset (NASA) [Tool]

'EDGE AI SMART FILTERING'



Edge ML Versioning & Artifact Traceability

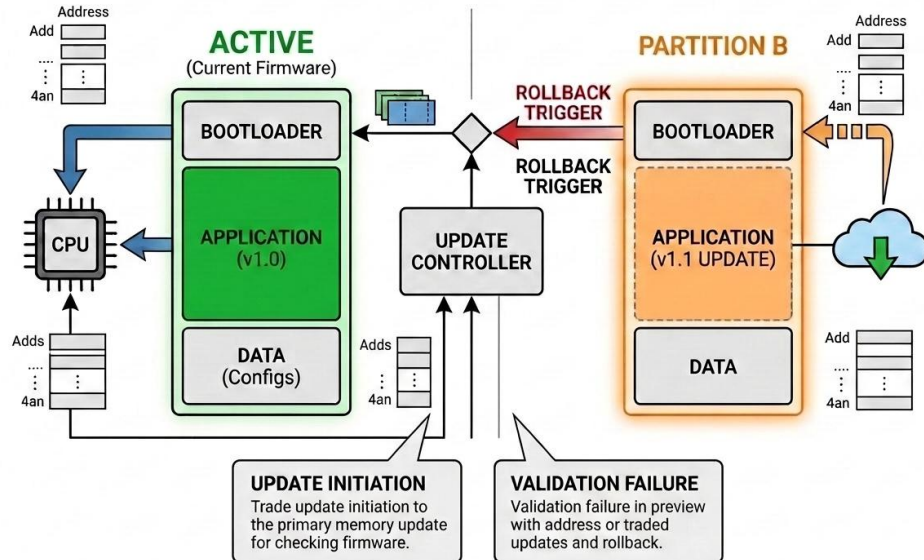
- The Inseparable Triad: Dataset + Code + Compiled Binary.
- Dataset Versioning with DVC [\[Tool\]](#)
- Model Registry & Tracking with MLflow [\[Tool\]](#)
- LiteRT FlatBuffer Model Metadata [\[Web\]](#)
- Key Elements: Binary SHA-256 hash, runtime compatibility, and quantization parameters (Scale/Zero-Point).



Over-The-Air (OTA) Updates & Resiliency

- Modular Updating: Replacing the .tflite artifact vs. Full OS images.
- Binary Diffs & Compression for constrained network channels.
- Dual A/B Partitions with Inference Watchdog:
 - * Autonomous rollback triggered by OOM errors or runtime crashes.
- IoT Fleet Management with Eclipse hawkBit [\[Web\]](#)

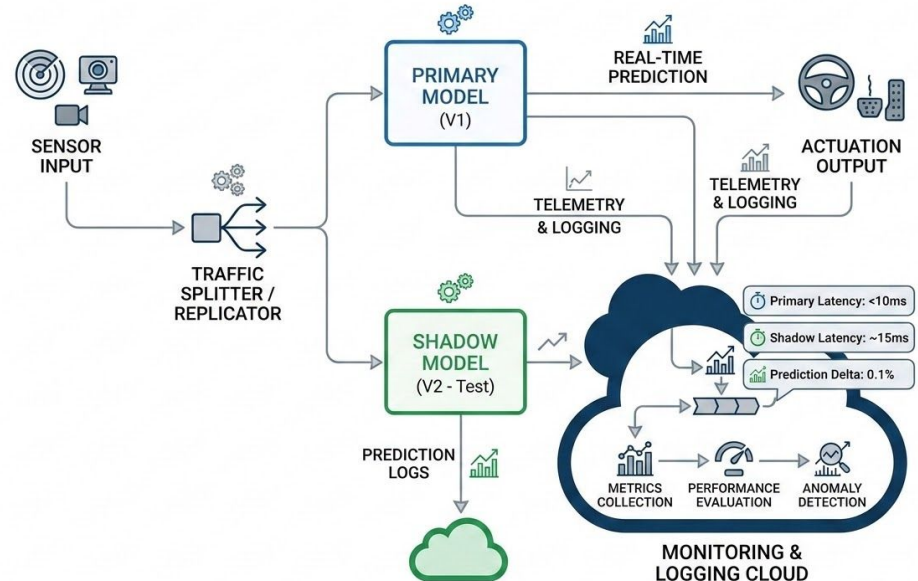
DUAL-BANK FLASH MEMORY PARTITION FOR SECURE FIRMWARE UPDATE & ROLLBACK



Production Deployment Patterns

- Industrial Risk Mitigation Strategies:
 - * Canary Deployment: Phased rollouts across 5% of edge nodes.
 - * Shadow Deployment (Silent Mode):
 - Legacy model retains direct control over physical actuators.
 - Candidate model runs in background on identical real-time inputs.
- Architecture Guide: MLOps Deployment Patterns (Google) [\[Web\]](#)

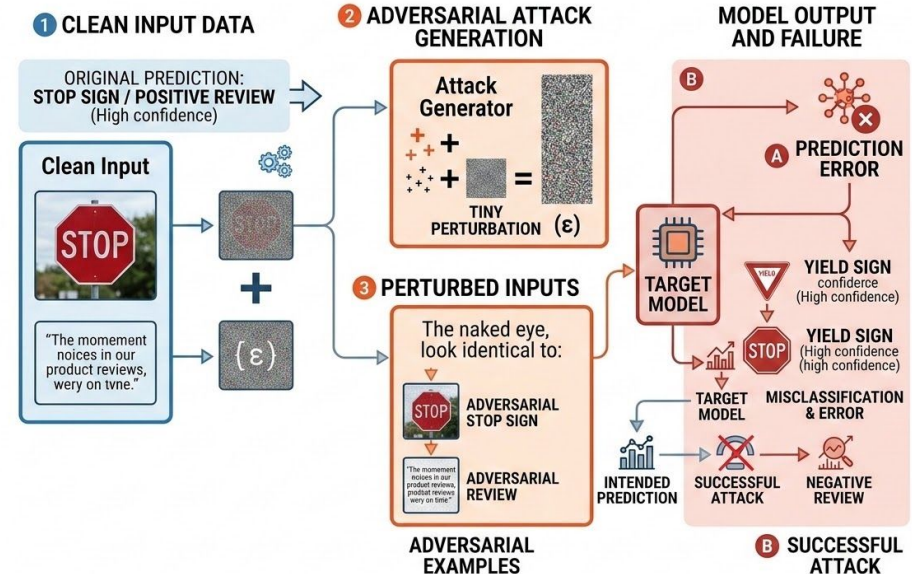
SHADOW DEPLOYMENT ARCHITECTURE



Edge Privacy, Security & Adversarial Risks

- Physical Threat Vector: Reverse-engineering model weights from flash storage.
- Vulnerability to Adversarial Attacks (Goodfellow et al.) [\[Paper\]](#)
 - * Imperceptible input noise forcing catastrophic misclassifications.
- Mandatory Defense Layers:
 - * Hardware Root of Trust & Secure Boot validation.
 - * Encrypted storage for LiteRT FlatBuffer artifacts at rest.

ADVERSARIAL MACHINE LEARNING ATTACK



Continuous Operation & Next Milestones

- The Edge ML Engineering Mindset:
 - * Statistical robustness + Thermal awareness + Artifact governance.
- Next Session (Thursday):
 - * From Evidence to Engineering Decision: Pareto Frontiers & Trade-offs.
- Saturday Test @ ISEP:

